

MOHAMMAD FARAJI
Machine Learning Researcher



🌐 farajimohammad.github.io
🌐 github.com/FarajiMohammad
📞 +98 930 244 2280 ✉️ mohammad.faraji@uok.ac.ir
📍 Representation Learning Lab (222), Computer Engineering Department, University of Kurdistan, Sanandaj, Iran

BRIEFLY

I'm a Machine Learning Researcher at the University of Kurdistan working on representation learning, Feature Selection, and Multi-Label learning. Our group is focused on fundamental methodological developments in supervised, semi-supervised, and unsupervised machine learning. Much of our work is related to representation models. I received my Master's in Artificial Intelligence from the Department of Computer, at University of Kurdistan in 2023. My supervisor was Dr. Fardin Akhlaghian and advisor was Amjad Seyed. My work mostly focused on the matrix factorization and low-rank approximation, predominantly for supervised multi-label feature selection and Multi-label learning problems. Before that, I obtained m Associate and Bachelor's degrees in Software Engineering.

RESEARCH INTERESTS

- Machine Learning, Deep Learning, Representation Learning
- Multi-Label Learning, Feature Selection, Optimization
- Numerical Linear Algebra, Low-Rank Approximation, Matrix Factorization

EXPERIENCE

M-Researcher Part-time	University of Kurdistan, SANANDAJ, IRAN, (Sep 2021 - Present) > skills : Problem solving, Python programming, Data Visualization, and MATLAB.
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EDUCATION

Master	Artificial Intelligence, UNIVERSITY OF KURDISTAN, SANANDAJ, IRAN, (Sep 2020 – Sep 2023) > Thesis title : Multi-Label Feature Selection with Global and Local Label correlation. > Supervisor : Dr. Fardin Akhlaghian. > Courses : machine learning, statistical pattern recognition, neural networks, advanced artificial intelligence, computer vision, digital image processing, complex Networks, and special subjects.
Bachelor	Software Engineering, SHAMSIPOUR TECHNICAL COLLEGE, TEHRAN, IRAN, (Sep 2017 – Jul 2019) > Project title : Student Management system.
Associate	Computer Software, YAZDANPANAHI TECHNICAL COLLEGE, SANANDAJ, IRAN, (Sep 2015 – Sep 2017) > Supplementary courses in computer science and software engineering.
TechSchool	Computer, TALEGHANI HIGH SCHOOL, SANANDAJ, IRAN, (Sep 2013 - Sep 2015) > A two-year education in basic computer science

PUBLICATION

2026	Robust log-based multi-label feature selection with dynamic label correlation and relevance-redundancy optimization M. Faraji, S. A. Seyed, and F. Akhlaghian. <i>Knowledge Based-Systems</i> , volume 342, 2026, pp.115825.
2024	Mutli-Label Feature Selection with Global and Local Label Correlation M. Faraji, S. A. Seyed, F. Akhlaghian, and R. Mahmoodi. <i>Expert Systems with Applications</i> , volume 246, 2024, pp.110179.

COMPUTER SKILLS

Operation Systems	Microsoft Windows and Linux
Word processing & Presentation	Office suites and $\text{\LaTeX}$
Vector and raster softwares	Adobe Illustrator and Abode XD
Development Tools	Pycharm, Jupyter Notebook, Colab, and Visual Studio
Web design	HTML, CSS, ASP.NET, and JavaScript

PROGRAMMING LANGUAGES

2021 – present **Python**, PyTorch, NumPy, Pandas, and scikit-learn
2022 – present **MATLAB**, linear algebra and visualization
2015 – 2017 **C++ | C#**, Software Engineering and Web development
2013 – 2015 **Basic | Visual Basic**, Software Engineering

PROJECT

Patter Recognition implemented several embedded feature selection algorithms, including methods based on label correlation analysis and robust optimization.
Image Processing Implemented image enhancement, segmentation, feature extraction, and object recognition algorithms using MATLAB.
Neural Networks Implemented associative memory and optimization algorithms based on Hopfield neural networks and analyzed convergence behavior.
Machine Learning Implemented classification, clustering, dimensionality reduction, and ensemble learning algorithms for benchmark datasets.

REFERENCES

Fardin Akhlaghian, *Associate Professor*

Department of Computer Engineering, University of Kurdistan, Sanandaj, Iran
Relationship : Master's Thesis Supervisor & Research Supervisor

@ f.akhlaghian@uok.ac.ir
☎ +98 918 873 8383

Amjad Seyedi, *Doctoral Researcher*

Department of Mathematics and Operational Research, Polytechnic Faculty of Mons, University of Mons, Belgium
Relationship : Master's Thesis Advisor

@ seyedamjad.seyedi@umons.ac.be
☎ +32 4 651 205 00

Parham Moradi, *Senior Lecturer*

College of Arts, Business, Law, Education and IT, Victoria University, Melbourne, Australia
Relationship : Former Course Instructor

@ parham.moradi@vu.edu.au
☎ +98 912 513 5478

Alireza Abdollahpouri, *Associate Professor*

Department of Computer Engineering, University of Kurdistan, Sanandaj, Iran
Relationship : Former Course Instructor

@ abdollahpouri@uok.ac.ir
☎ +98 918 877 0993

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